

Original Software Publication

Herschel vision: A hyperspectral image processing software for data preparation in machine learning pipelines

Billy G. Ram, Sunil GC, Xin Sun^{*}

Department of Agricultural and Biosystems Engineering, North Dakota State University, Fargo, ND 58102, USA

ARTICLE INFO

Keywords:

hyperspectral analysis
data preparation
Python
GUI
application

ABSTRACT

In recent years, publications have relied on either commercial software or custom-developed codes for the analysis of hyperspectral images. This practice has inadvertently restricted the broader adoption of hyperspectral imaging. Herschel Vision is an open-source hyperspectral image analysis application that enables the users to visualize and analyze hyperspectral images. Herschel Vision can automate white and dark reference calibration, spectral data visualization and extraction, generation of pseudo RGB images, cropping and segmentation of region of interests, saving images as three-dimensional cubes, and applying forward feed signal pre-processing corrections. Herschel Vision can handle proximally recorded hyperspectral data for curation of machine learning and deep learning training datasets.

Metadata

Nr	Code metadata description	Metadata
C1	Current code version	v1.0.0
C2	Permanent link to code/repository used for this code version	https://github.com/billygrahamram/HerschelVision.git
C3	Legal code license	GPL-2.0 license
C4	Code versioning system used	git
C5	Compilation requirements, operating environments and dependencies	OS: Windows, Mac OS, Ubuntu. Dependencies: requirements.txt file provided.
C6	Software code languages, tools and services used	PlantCV, Python 3.10, CustomTkinter
C7	Support for questions	Github Discussions: https://github.com/billygrahamram/HerschelVision/discussions

1. Motivation and significance

Hyperspectral imaging offers comprehensive chemometric and spectroscopic information on biotic and abiotic materials [1]. Recent advancements in hardware technology have led to the development of handheld proximal hyperspectral sensors that are capable of recording high spatial and spectral resolution images [2]. While remote sensing applications primarily concentrate on the creation of detailed maps of

the earth's surface, the emphasis has shifted with proximal sensing [3–5]. The latter is more geared towards the development and training of machine learning (ML) and deep learning (DL) algorithms for in-field applications [6]. The growing emphasis on in-field applications has also transformed existing analysis methods. Instead of relying on a limited number of large ortho tiles recorded from orbital and suborbital platforms, researchers now utilize extensive, carefully curated datasets comprising of in-situ images.

Object detection and classification tasks require large training datasets. Curation of such dataset involves the selection of region of interest (ROI) from raw images. High adaptability of RGB images has resulted in several commercial and open-source software that allow image cropping and bounding box based ROI selection. But hyperspectral imaging has not been equally adapted, resulting in very few applications that aid in hyperspectral data preparation.

Unfortunately, the majority of hyperspectral data preparation tools are either proprietary to specific sensor manufacturer, or commercial, limiting access and innovation in this area. Hyperspectral data captures wide range of electromagnetic radiation, ranging from visible near infrared to shortwave infrared (400–2500 nm). Recording hundreds of wavelengths, renders a three-dimensional (x, y, z) cube as the output image, inherently increasing the dimensionality of the data, and thereby affecting its size, complexity, and computational cost. The three-dimensional structure of hyperspectral data can be understood by dividing it into three dimensions: x and y represent the spatial

^{*} Corresponding author.

E-mail address: xin.sun@ndsu.edu (X. Sun).

<https://doi.org/10.1016/j.softx.2025.102089>

Received 10 December 2024; Received in revised form 2 February 2025; Accepted 8 February 2025

Available online 14 February 2025

2352-7110/© 2025 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

coordinates, while z represents the spectral dimension.

Proximal hyperspectral data can be used in various forms: one-dimensional spectroscopic studies utilize only the spectral information, while two-dimensional analysis focuses on individual spatial channels. Three-dimensional analysis, however, combines both spatial and spectral data. Depending on the type of analysis, specific data pipelines are undertaken. In one-dimensional spectroscopic studies, only the spectral information is utilized, with spatial information removed through pre-processing methods. These studies are dependent on signal correction through pre-processing methods; like derivatives for signal smoothing, scatter correction and normalization [7]. Whereas studies using spatial and spectral data utilize the three-dimensional hyperspectral cube. These studies utilize segmentation (removal of background) and ROI selection to select important areas in the image cube [8].

There is an increasing need in the hyperspectral imaging community for an open-source graphical user interface (GUI) application that can expedite the data preparation of proximal hyperspectral data for ML and DL analysis. Such an application can expedite data visualization and curation, resulting in faster model development and evaluation. The objective of this study is to introduce Herschel Vision, an open-source, GUI application capable of hyperspectral data visualization, spectral data pre-processing, segmentation, ROI selection and unfolding of 3D data.

2. Software description

2.1. Overview

Herschel Vision, (Version 1.0.0), is written using Python 3.10 and utilizes CustomTkinter library for its GUI. The application features an intuitive GUI that is designed to be accessible to both novice and experienced users. The application adheres to the Model-View-Controller (MVC) design pattern, which separates code into distinct components for efficient and scalable development. The application allows for over the cloud updates (through Github) to ensure future compatibility with new analysis pipelines and data types. The application is designed to work seamlessly with a variety of well-known proximal hyperspectral camera image formats, including ENVI's .raw files. The application facilitates automated reference calibration if the white (named as WHITEREF) and dark (named as DARKREF) reference images are found in the same directory as the hyperspectral image Fig. 1.

2.2. Data visualization

Upon launching the application, users are presented with the data

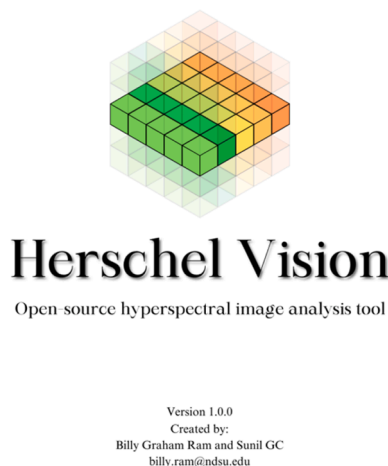


Fig. 1. Herschel Vision about page.

visualization window (Fig. 2), where they can visualize hyperspectral image channels, plot the spectral curve of a selected point using the cursor, and view a preliminary scatter plot composed of two clusters (with the ability to adjust the number of clusters in the preference window). This window also includes sliders that enable viewing each channel of the hyperspectral image, allowing the users to visualize sensor noise (usually associated with initial channels), and sliders that allow users to select the x and y channels for the scatter plot. The primary objective of the data visualization window is to provide an initial overview of the hyperspectral image data.

2.3. Pre-processing

In the pre-processing window, users can select and apply individual or sequential pre-processing filters in a forward-feed approach (Fig. 3). This window also allows visualization of raw spectral plot and pre-processed spectral plot for easy understanding of a filter's effect on the raw data. These pre-processed spectral signatures can be saved as tables (.csv) using the "save pre-processed data" button. The application offers a range of pre-processing techniques, including the Standard Normal Variate (SNV) method, Savitzky-Golay filters for calculating derivatives, and normalization to scale raw values between 0 and 1.

2.4. Cropping and segmentation

In the cropping and segmentation window, users can define a ROI by selecting a bounding box within the hyperspectral image (Fig. 4). After defining the ROI, users can employ segmentation techniques to separate the foreground from the background. The application employs K-means clustering to generate a segmentation map, which enables visualization of clusters and facilitates selection of areas to remove. Users can save their selected ROI or segmented ROI as 3D cubes or unfolded tables using the provided buttons.

2.5. Preferences

The application offers a comprehensive preferences menu, allowing users to customize various parameters and tailor the data processing stages to their requirements. Users can fine-tune Savitzky-Golay filter settings by adjusting the window size (default: 15) and polynomial order. The number of clusters in the segmentation algorithm can be adjusted from the default value of 2 to better suit the specific characteristics of the hyperspectral data. Users can choose the number of RGB bands to display a pseudo-color representation of the hyperspectral data, enhancing visual understanding and exploration. These options are available to give the user much freedom over the data processing stages.

3. Illustrative examples

3.1. Dataset

To test the underworking of the application, Herschel Vision was used to prepare the input data for a one-dimensional convolutional neural network (1D-CNN) and a three-dimensional convolutional neural network (3D-CNN) classification model. WeedCube, an open-source proximal hyperspectral weed dataset, was used in this study. It is freely accessible at <https://doi.org/10.15482/USDA.ADC/25306255.v1> [9,10]. The dataset consists of four weeds (kochia, redroot pigweed, and waterhemp, ragweed) and three crops (canola, sugarbeet, and soybean). Each image in the dataset is reference calibrated and consists of 224 channels. For validating the application, 1D-CNN and 3D-CNN models were trained for classification of three weeds (kochia, ragweed, and waterhemp).

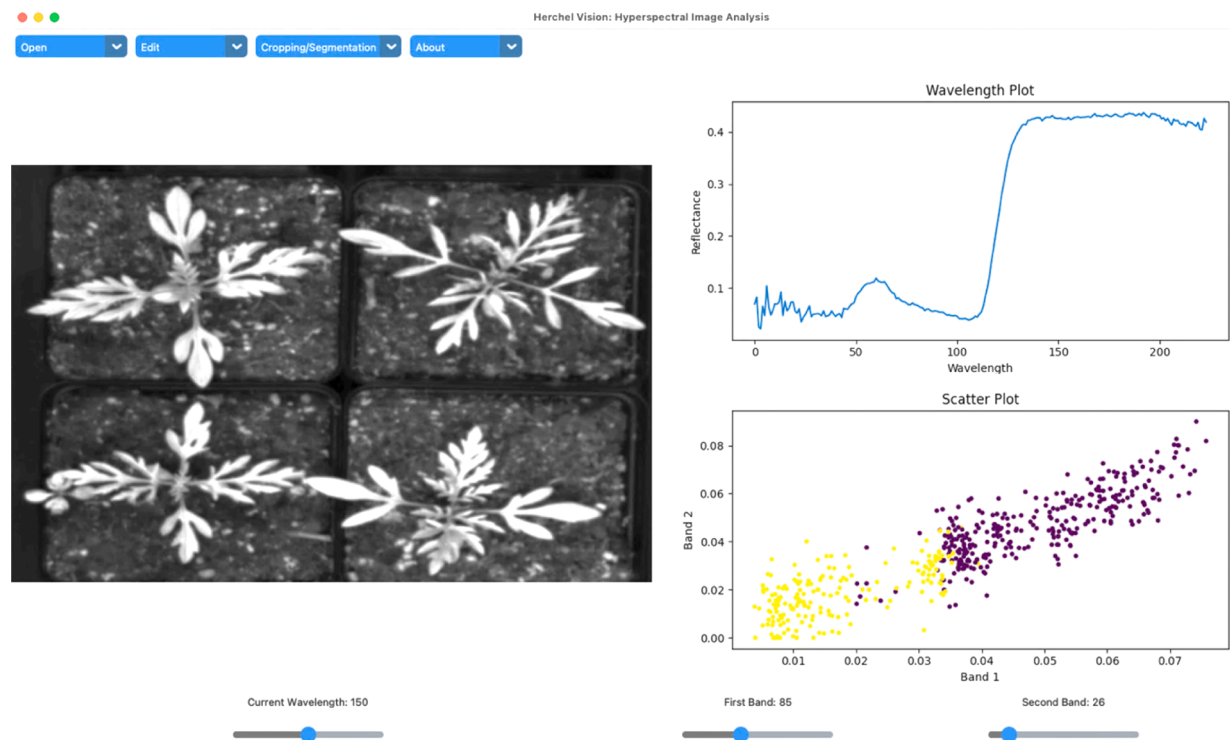


Fig. 2. The home window of Herschel Vision. User selected points are plotted on the right under Wavelength Plot. A Scatter Plot of two clusters against any bands in displayed below. The users have the option to visualize any channel from the input hyperspectral data.

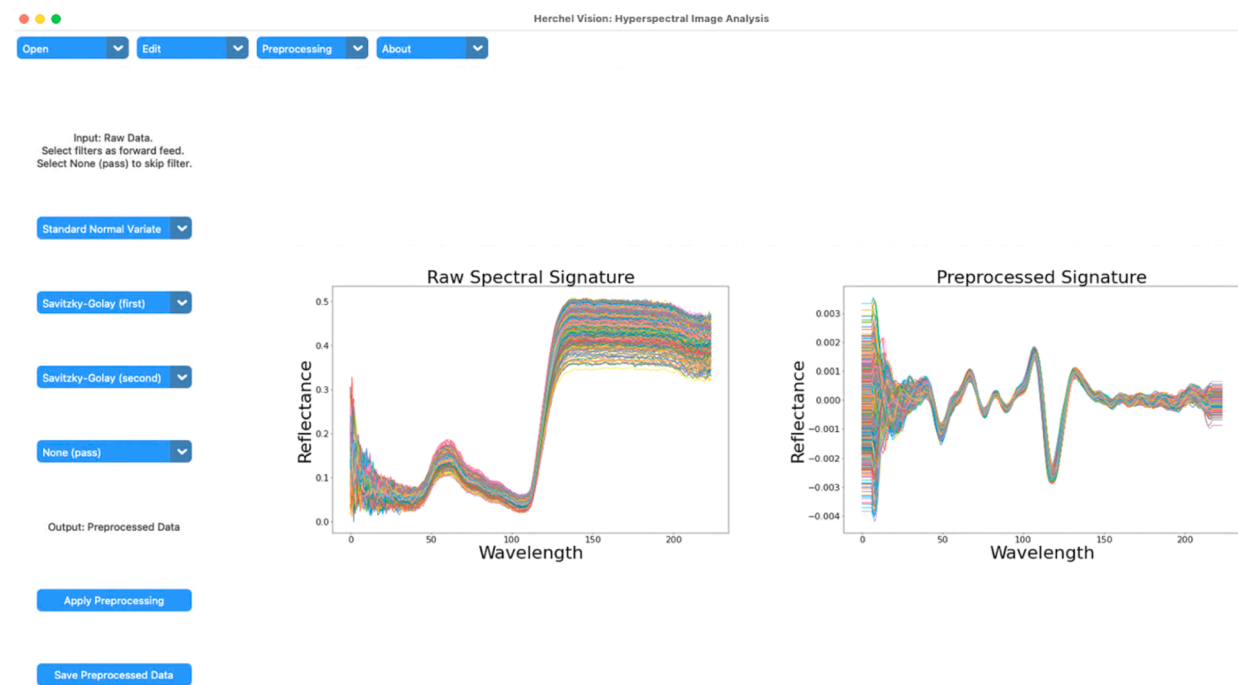


Fig. 3. Herschel Vision spectral signature pre-processing window.

3.2. Selection of region of interest and point spectral signature

The application’s cropping and segmentation window can be accessed by selecting the ‘Tools’ option from the main menu. Once opened, the imported image is visible on the left side window. The user can select multiple ROIs from a single image. The dimension of the bounding box is assigned by the user’s cursor movement and can be of

any length or width. Allowing the user to select whole objects in the ROI (Fig. 5) or a very small area (Fig. 6).

3.3. Data export for analysis

The cropping tool was used to extract ROIs from whole plant images (Fig. 5), which were then saved as NumPy arrays for further processing.

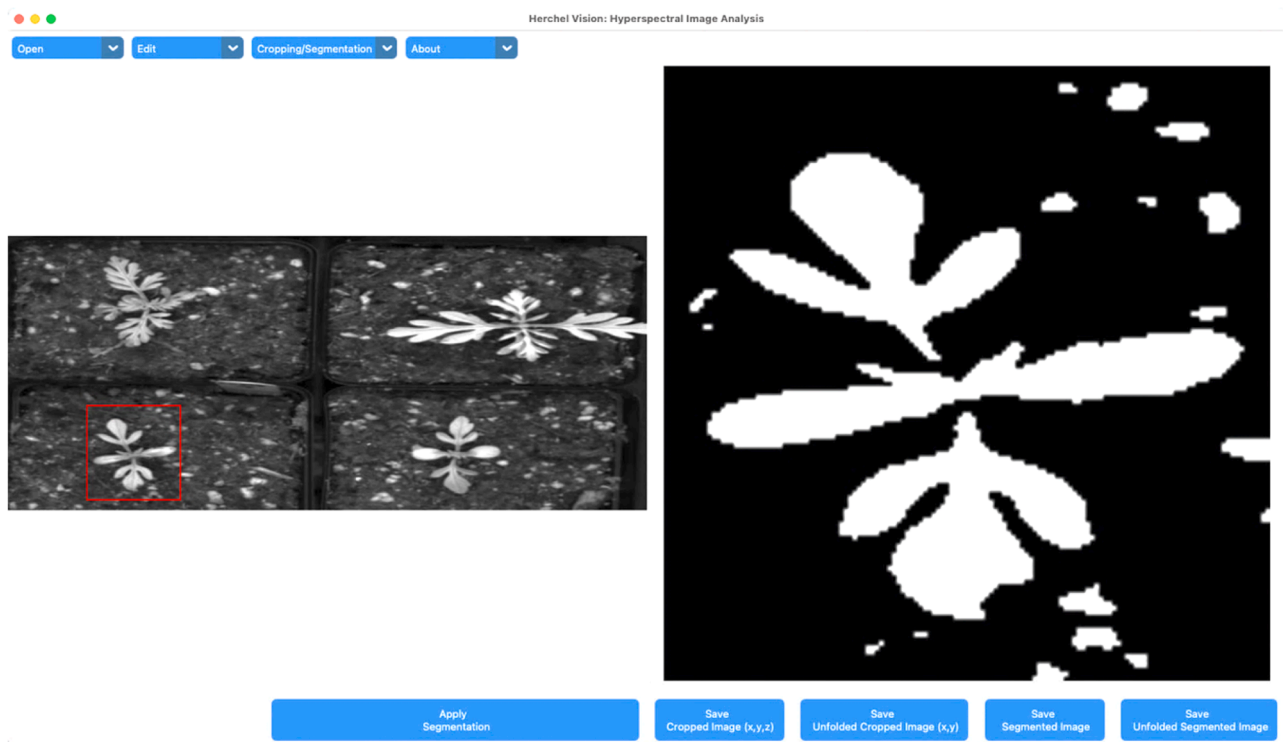


Fig. 4. Herschel Vision Segmentation window.

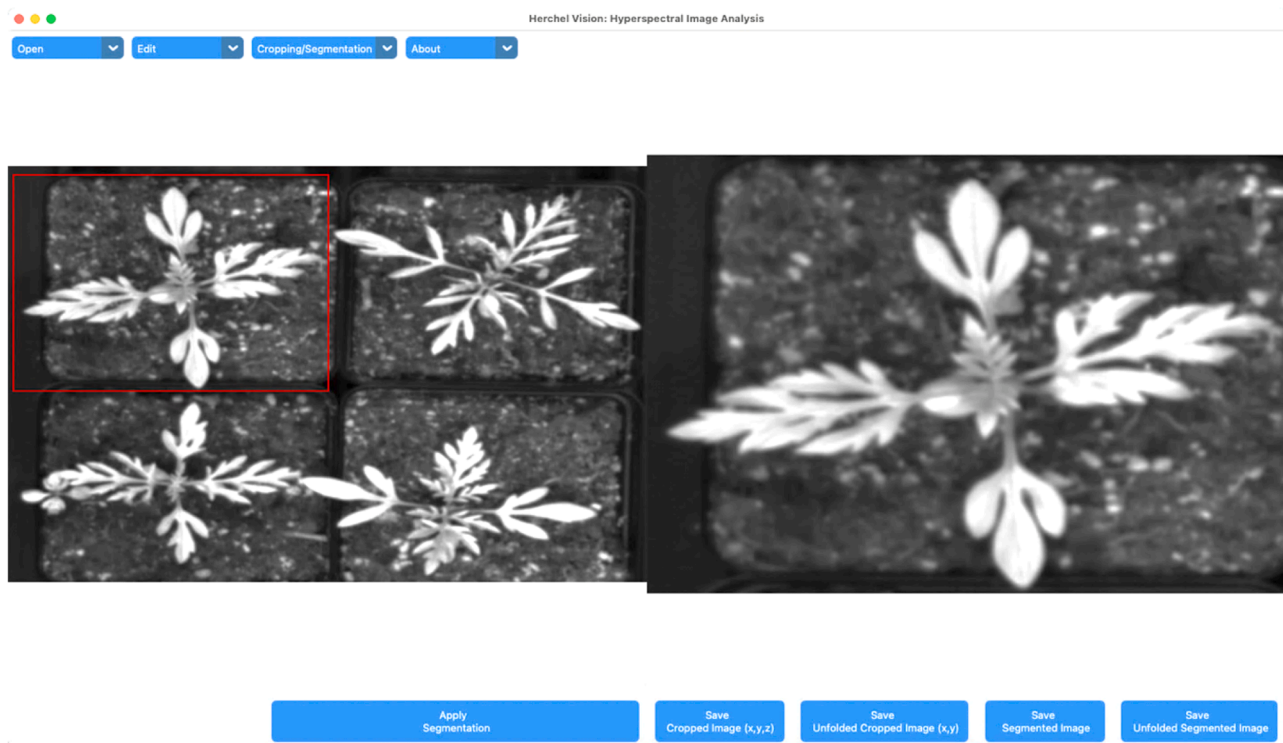


Fig. 5. The cropping and segmentation window of Herschel Vision application. User selected ROI (red box) is displayed on the right window. The buttons below allow the user to save the data.

These files were used for 3D-CNN model training. Multiple smaller ROI (Fig. 6) from different leaf area were cropped and saved as “Unfolded Cropped Image (x, y)”. These were saved in the directory as .csv files of multiple rows and 224 columns. These table files were used to train the 1D-CNN model.

3.4. Data preparation and deep learning training

Following export from the Herschel Vision application, the data was separated into individual directories for 1D-CNN and 3D-CNN model training purposes. The data were then imported into a Python script

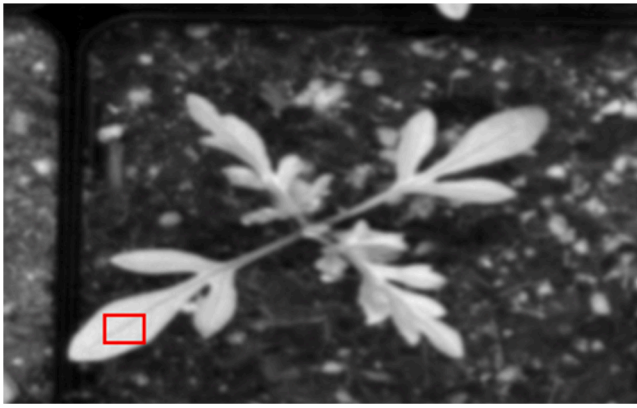


Fig. 6. Example of a smaller ROI selection. This is useful in selecting only the pure spectra of a material.

where it was augmented, resized and used for training. To reduce the dimensionality of the hyperspectral data, 54 channels out of the 224 channels were selected. Dimensionality reduction was implemented by selecting every third channel starting from the 62nd channel till the last 224th channel. 5000 spectra per class were selected for 1D-CNN model. Pre-processed data using Savitzky-Golay 1st derivative, Savitzky-Golay 2nd derivative, and Standard Normal Variate, were stacked horizontally to the raw data. Data for 3D-CNN was resized to $100 \times 100 \times 54$, and five data augmentation (transpose, normalized, flipping across all dimensions, vertical flip, and horizontal flip) techniques were applied to the raw image. With augmentation, the number of training samples increased to 300 images per class. The model training achieved good accuracy of 99.7 and 97.7 for 1D-CNN and 3D-CNN models respectively (Fig. 7).

3.5. Limitations and future updates

The compatibility of the application and its dependencies was evaluated on Mac, Windows, and Linux (Ubuntu) operating systems, confirming its operational functionality. Identified limitations include the lack of support for batch image processing, DL-based segmentation, and morphological masking, as well as a startup delay. These limitations will

be addressed in future updates as part of the development roadmap, with version 5 expected to resolve them.

4. Impact

Satellite-based hyperspectral imaging analysis is supported by several commercially available GIS software, including ArcGIS (Environmental Systems Research Institute Inc., Redlands, CA, USA) and ENVI (Exelis Visual Information Solutions, Boulder, CO, USA). While these tools are primarily designed for satellite imagery, they can also be used for proximal hyperspectral data visualization and analysis due to their similar data structures. However, the exponentially higher dimensionality of proximal hyperspectral data compared to satellite data presents computational challenges, making GIS software less efficient for large datasets. Proximal hyperspectral data has been widely utilized in research across agriculture, food science, medicine, geology, and civil engineering. With the increasing emphasis on ML and DL models in data analysis pipelines, new applications focusing on proximal hyperspectral data have emerged in recent years. Table 1 presents a list of such applications. A common trend among recently developed tools is their reliance on commercial software such as MATLAB [11,12]. Free standalone applications often lack intuitive GUIs or have ceased updates, limiting their compatibility with modern operating systems and reducing performance efficiency [13,14]. While commercial software effectively addresses modern ML and DL analysis needs, their high cost remains a significant barrier [15]. Herschel Vision aims to address these challenges by providing an intuitive GUI with support for major data formats, ensuring accessibility and efficiency in proximal hyperspectral data analysis.

5. Conclusions

The application of proximal hyperspectral imaging *in-situ* research is steadily increasing. To enhance its adoption across various scientific disciplines, it is essential to develop user-friendly tools that facilitate hyperspectral data analysis, particularly for first-time users. Herschel Vision is a user-friendly, GUI-based application designed for this purpose. It is compatible with the latest versions of Windows, Mac, and Linux operating systems and supports data from commercially available proximal hyperspectral sensors. The software enables users to visualize

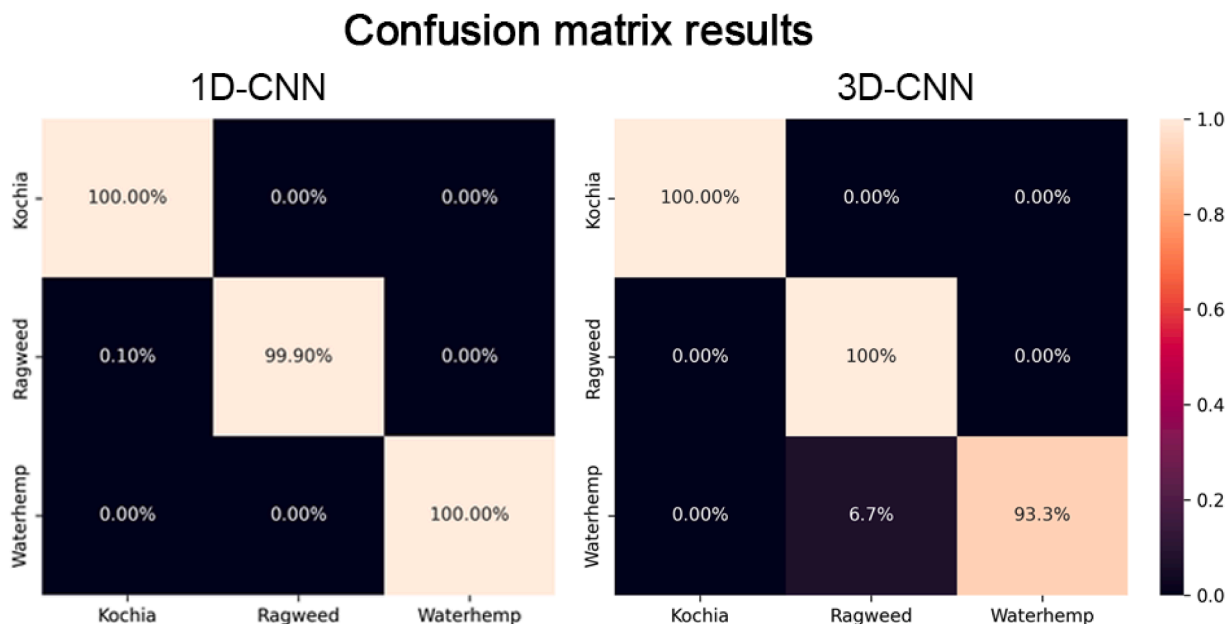


Fig. 7. Confusion matrix plots for 1D-CNN and 3D-CNN models. Showcasing good performance against false positives and negatives.

Table 1

Some applications, toolboxes, and packages that allow analysis of hyperspectral image.

Application	License	Last Update	Dependency	Operating System	GUI	Remark	Reference
HIA Toolbox	Paid*	2004	MATLAB	/	Yes	Application has not been updated in a decade and therefore is not compatible with newer versions of MATLAB.	Arzuaga-Cruz, Jimenez-Rodriguez [11]
HyperCube	Free	2015	Standalone	Mac OS PowerPC & Windows 7	Yes	Application has not been updated recently and is not compatible with latest versions of operating systems. Therefore, it's functionality cannot be tested.	ERDC [16]
Spectral Python	Free	2020	Python	/	No	A scripting-based Python library for analyzing hyperspectral data. Limitations include the requirement for experience with Python programming.	Thomas Boggs, Don March [14]
Hyper-Tools Toolbox	Paid*	2020	MATLAB	/	Yes	Allows ROI selection, segmentation, pre-processing, regression, classification. Limitations include dependency on MATLAB data formats. That require some programming knowledge for image input and output. Model training is limited to a single image application.	Mobaraki and Amigo [12]
HSDAR R package	Free	2022	R	/	No	Scripting based package geared towards spectral analysis. Limitations include no support for ROI selection, segmentation, and pre-processing.	Lehnert, Meyer [17]
Scyllarus (Scyven)	Free	2023	Standalone	Windows 7 & Ubuntu	Yes	Application allows visualization of hyperspectral images, plot spectral plot, and apply clustering. Limitation includes limited support for latest versions of operating system. Resulting in slow performance.	Habili and Oorloff [13]
Prediktera (Evince)	Paid	2024	Standalone	Windows, Mac, & Ubuntu	Yes	Application allows an intuitive GUI based hyperspectral image processing. Limitation includes a hefty commercial price.	Prediktera [15]
HerschelVision	Free	2024	Standalone	Windows, Mac, & Ubuntu	Yes	Application allows hyperspectral image visualization, spectral plotting, scatter plot, ROI selection, segmentation, pre-processing and unfolding of spectral data.	This paper

* Free package or toolbox that is compatible with commercially available software.
/ Compatibility depends on parent software

hyperspectral data cubes, extract and plot reflectance spectra by selecting points in an image and generate scatter plots of prominent spectral clusters. Additionally, it provides functionalities for cropping, selecting regions of interest (ROI), background removal, and data pre-processing. Herschel Vision allows users to save processed data in multiple formats, including 3D cube files (.npy) and unfolded tabular data (.csv). As an open-source application, it supports continuous development and community contributions, ensuring ongoing improvements and expanded capabilities.

CRediT authorship contribution statement

Billy G. Ram: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Sunil GC:** Writing – review & editing, Software, Methodology. **Xin Sun:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

No financial or non-financial interest in the subject matter or materials discussed in this manuscript.

Acknowledgements

This material is based upon work partially supported by the North Dakota State Department of Agriculture, FAR0037869. This work is/was supported by the USDA National Institute of Food and Agriculture, Hatch project number ND01487.

References

- [1] Ram BG, et al. A systematic review of hyperspectral imaging in precision agriculture: analysis of its current state and future prospects. *Comput Electron Agric* 2024;222:109037. <https://doi.org/10.1016/j.compag.2024.109037>.
- [2] Behmann J, et al. Specim IQ: evaluation of a new, miniaturized handheld hyperspectral camera and its application for plant phenotyping and disease

detection. *Sensors* (Basel, Switzerland) 2018;18. <https://doi.org/10.3390/s18020441>.

- [3] Ram BG, et al. Palmer amaranth identification using hyperspectral imaging and machine learning technologies in soybean field. *Comput Electron Agric* 2023;215:108444. <https://doi.org/10.1016/j.compag.2023.108444>.
- [4] Aryal S, Chen Z, Tang S. Mobile hyperspectral imaging for material surface damage detection. *J Comput Civ Eng* 2021;35(1):04020057. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000934](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000934).
- [5] Fei, B., Chapter 3.6 - hyperspectral imaging in medical applications, in *Data handling in science and technology*, J.M. Amigo, Editor. 2019, Elsevier. p. 523–565 DOI: <https://doi.org/10.1016/B978-0-444-63977-6.00021-3>.
- [6] Mishra P, et al. Close-range hyperspectral imaging of whole plants for digital phenotyping: recent applications and illumination correction approaches. *Comput Electron Agric* 2020;178. <https://doi.org/10.1016/j.compag.2020.105780>.
- [7] Pavia, D.L., et al., Introduction to spectroscopy. 2014: cengage learning DOI: <https://doi.org/10.1021/ed056pa323.2>.
- [8] Eddy P, et al. Hybrid segmentation-artificial neural network classification of high resolution hyperspectral imagery for site-specific herbicide management in agriculture. *PE&RS* 2008;74(10):1249–57. <https://doi.org/10.14358/PERS.74.10.1249>.
- [9] Ram BG, et al. WeedCube: proximal hyperspectral image dataset of crops and weeds for machine learning applications. *Data in Brief* 2024;56:110837. <https://doi.org/10.1016/j.dib.2024.110837>.
- [10] Ram, B.G., et al., Proximal hyperspectral image dataset of various crops and weeds for classification via machine learning and Deep learning techniques. 2024 DOI: <https://doi.org/10.15482/USDA.ADC/25306255.v1>.
- [11] Arzuaga-Cruz E, et al. A MATLAB toolbox for hyperspectral image analysis. In: *IGARSS 2004. 2004 IEEE International Geoscience and Remote Sensing Symposium*; 2004. <https://doi.org/10.1109/IGARSS.2004.1370246>.
- [12] Mobaraki N, Amigo JM. HYPER-Tools. A graphical user-friendly interface for hyperspectral image analysis. *Chemometrics Intellig Lab Syst* 2018;172:174–87. <https://doi.org/10.1016/j.chemolab.2017.11.003>.
- [13] Habili N, Oorloff J. Scyllarus™: from research to commercial software. In: *Proceedings of the ASWEC 2015 24th Australasian software engineering conference*; 2015. <https://doi.org/10.1145/2811681.2817752>.
- [14] Thomas Boggs, et al., Spectral python (SPy) 0.23.1, Zenodo, Editor. 2022, Zenodo DOI: <http://doi.org/10.5281/zenodo.7135091>.
- [15] Prediktera. Evince. Prediktera 2024; Available from: <https://prediktera.com>.
- [16] ERDC, G.R.L. HyperCube: spectral imagery analysis software. US Army Corps of Engineers; 2015. Available from, <https://www.erdcc.usace.army.mil/media/fact-sheets/fact-sheet-article-view/article/610433/hypercube/>.
- [17] Lehnert, L.W., et al., Hyperspectral data analysis in R: the hsdar package. *ArXiv preprint arXiv:1805.05090*, 2018 DOI: <https://doi.org/10.48550/arXiv.1805.05090>.